PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 7, 1995

The transfer of a chemical from one phase to another is an important process. For phase equilibrium, when a compound A is distributed between two phases at constant T and P, we can write for its chemical potential, μ_A:

$$\mu_A(\text{phase 1}) = \mu_A(\text{phase 2})$$

where $\mu_A = \underline{G}_A$, and $\underline{G}_A$ is the partial molal free energy.

Since the chemical potential of A depends on a_A, the activity of A, i.e.,

$$\mu_A = \mu_A^0 + RT \ln a_A$$

its activity in the two phases will be the same at equilibrium, if the same standard state is used for A in the two different phases.

In living biological cells the concentration of sodium ions inside the cell is kept at a lower concentration than the concentration outside the cell, because sodium ions are actively transported from the cell.

a) Consider the following process at 37°C and 1atm, where *M* is mol/lit.

1 mol NaCl(0.05 *M* inside) → 1 mol NaCl (0.20 *M* outside)

Write an expression for the free energy change, $\Delta\underline{G}$, for this process in terms of activities. Define all symbols used. Assume that the standard state for NaCl is the same inside and outside.

b) Calculate $\Delta\underline{G}$ for the process. You may approximate the activities by concentrations in *M*. Will the process proceed spontaneously? R=8.314 J K^{-1} mol^{-1}.

c) Calculate $\Delta\underline{G}$ for the process at equilibrium.

d) The standard free energy for hydrolysis of ATP to ADP (ATP + H_2O → ADP + phosphate) in solution is ΔG^0 = -31 kJ/mol at 37°C and 1atm. The free energy of this reaction can be used to power the sodium-ion pump. For a ratio of ATP to ADP of 10, what must be the concentration of phosphate to obtain -40kJ/mol for the hydrolysis? Assume that the activity coefficients are 1 for the calculation.